

Weekly Progress Report

Mathematical Study of Radar Scattering from Heterogeneous Rough Soil

Week 1

1 Problem

The objective is to study electromagnetic scattering from a rough soil surface whose dielectric properties may vary with depth. The motivation is radar-based sensing of soil moisture. The paper models the soil surface as a two-dimensional rough interface

$$z = f(x),$$

and studies the scattered electromagnetic field produced when a radar wave is incident on the surface.

The main physical relationship is

soil moisture $\longrightarrow$ dielectric properties $\longrightarrow$ electromagnetic scattering.

Let m_v denote volumetric soil moisture and ϵ_r the relative permittivity. The paper uses a moisture-dependent dielectric model

$$\epsilon_r = \epsilon_r(m_v).$$

Hence, for a depth-dependent moisture profile,

$$m_v = m_v(z) \quad \Rightarrow \quad \epsilon_r = \epsilon_r(z).$$

Thus, the electromagnetic material properties are spatially varying.

2 Electromagnetic Formulation

The electromagnetic fields are governed by Maxwell's equations. Under a time-harmonic convention,

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}, \quad \nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}.$$

These equations can be reduced to a wave equation for the relevant field U :

$$\nabla \times \left(\frac{1}{p} \nabla \times U \right) - k_0^2 q U = 0,$$

where k_0 is the free-space wavenumber.

For TM polarization,

$$U = H_y, \quad p = \epsilon_r, \quad q = \mu_r,$$

while for TE polarization,

$$U = E_y, \quad p = \mu_r, \quad q = \epsilon_r.$$

Because the surface is rough and ϵ_r may depend on position, obtaining an analytical solution is difficult. A numerical method is therefore required.

3 Finite Element Method

The finite element method (FEM) converts the continuous boundary-value problem into a finite-dimensional approximation.

The computational domain Ω is divided into a finite number of small elements. For the problem considered here, triangular elements are used:

$$\Omega = \bigcup_{e=1}^{N_e} \Omega_e.$$

Instead of finding the exact continuous field $U(\mathbf{r})$, we approximate it using basis functions:

$$U_h(\mathbf{r}) = \sum_{i=1}^N u_i N_i(\mathbf{r}),$$

where N_i are basis functions and u_i are unknown coefficients.

Therefore, FEM transforms the problem

find a continuous function $U(\mathbf{r})$

into

find the finite vector $\mathbf{u} = [u_1, u_2, \dots, u_N]^T$.

4 Weak Formulation

Let T be a suitable test function. Instead of requiring the wave equation to hold pointwise, FEM uses a weighted integral formulation.

For the electromagnetic problem, the weak form used in the paper is

$$\int_{\Omega} \left[(\nabla \times T) \cdot \left(\frac{1}{p} \nabla \times U \right) - k_0^2 q T \cdot U \right] dS = \text{boundary terms.}$$

The weak formulation is obtained by multiplying the differential equation by a test function, integrating over the domain, and applying the appropriate vector identities and boundary terms.

5 Galerkin Discretization

The FEM approximation

$$U_h = \sum_{i=1}^N u_i N_i$$

is substituted into the weak form.

In the Galerkin method, the test functions are chosen from the same finite-dimensional space as the basis functions. Taking $T = N_j$ gives one equation for each j :

$$\sum_{i=1}^N u_i \int_{\Omega} \left[(\nabla \times N_j) \cdot \left(\frac{1}{p} \nabla \times N_i \right) - k_0^2 q N_j \cdot N_i \right] dS = b_j.$$

Hence the continuous PDE is converted into the linear system

$$\boxed{A\mathbf{u} = \mathbf{b}}.$$

This is the main computational form of the FEM problem.

6 Element Matrices and Assembly

The integrals are evaluated separately over each element Ω_e . This gives a local element matrix

$$A^{(e)}.$$

The local matrices are then combined according to the connectivity of the mesh to form the global matrix

$$A.$$

Thus,

$$\boxed{\text{element equations} \rightarrow \text{element matrices} \rightarrow \text{global system.}}$$

Since the basis functions are local, each degree of freedom interacts only with nearby degrees of freedom. Consequently, the global matrix is sparse.

The paper uses first-order Whitney elements for the electromagnetic field representation and solves the resulting sparse system using a sparse direct solver.

7 Boundary Conditions

The physical problem is unbounded, but numerical computation requires a finite domain. Artificial boundaries are therefore introduced.

A simple boundary would cause artificial reflection of electromagnetic waves. The paper uses absorbing boundary conditions (ABCs) to approximate an outgoing-wave condition so that the scattered field can leave the computational domain with minimal reflection.

The boundary condition contributes additional terms to the weak formulation and hence to the final matrix system.

8 Current Stage of the Work

The work completed so far has focused on understanding the mathematical structure of the problem:

$$\boxed{m_v(z) \rightarrow \epsilon_r(z) \rightarrow \text{Maxwell's equations} \rightarrow \text{wave equation} \rightarrow \text{weak form} \rightarrow \text{FEM discretization} \rightarrow A\mathbf{u} = \mathbf{b}}$$

The next stage will be to derive the weak formulation and element matrices in greater detail, understand the Whitney basis functions and their role in electromagnetic FEM, and then study how the numerical solution is used to obtain the scattered far field and radar cross section.